

$\therefore +$ is an associative operator here.

$$\begin{aligned} \bullet 0 \cdot (0 \cdot 0) &= 0 \cdot 0 = 0 & \bullet 0 \cdot (0 \cdot 1) &= 0 \cdot 1 = 1 \\ (0 \cdot 0) \cdot 0 &= 0 \cdot 0 = 0 & (0 \cdot 0) \cdot 1 &= 0 \cdot 1 = 1 \end{aligned}$$

$$\begin{aligned} \bullet 0 \cdot (1 \cdot 0) &= 0 \cdot 1 = 1 & \bullet 0 \cdot (1 \cdot 1) &= 0 \cdot 1 = 1 \\ (0 \cdot 1) \cdot 0 &= 1 \cdot 0 = 1 & (0 \cdot 1) \cdot 1 &= 1 \cdot 1 = 1 \end{aligned}$$

$$\begin{aligned} \bullet 0 \cdot (0 \cdot 2) &= 0 \cdot 2 = 2 & \bullet 0 \cdot (1 \cdot 2) &= 0 \cdot 2 = 2 \\ (0 \cdot 0) \cdot 2 &= 0 \cdot 2 = 2 & (0 \cdot 1) \cdot 2 &= 1 \cdot 2 = 2 \end{aligned}$$

$$\begin{aligned} \bullet 0 \cdot (2 \cdot 0) &= 0 \cdot 2 = 2 & \bullet 0 \cdot (2 \cdot 1) &= 0 \cdot 2 = 2 \\ (0 \cdot 2) \cdot 0 &= 2 \cdot 0 = 2 & (0 \cdot 2) \cdot 1 &= 2 \cdot 1 = 2 \end{aligned}$$

$$\begin{aligned} \bullet 0 \cdot (2 \cdot 2) &= 0 \cdot 2 = 2 & \bullet 1 \cdot (0 \cdot 0) &= 1 \cdot 0 = 1 \\ (0 \cdot 2) \cdot 2 &= 2 \cdot 2 = 2 & (1 \cdot 0) \cdot 0 &= 1 \cdot 0 = 1 \end{aligned}$$

$$\begin{aligned} \bullet 1 \cdot (0 \cdot 1) &= 1 \cdot 1 = 1 & \bullet 1 \cdot (0 \cdot 2) &= 1 \cdot 2 = 2 \\ (1 \cdot 0) \cdot 1 &= 1 \cdot 1 = 1 & (1 \cdot 0) \cdot 2 &= 1 \cdot 2 = 2 \end{aligned}$$

$$\begin{aligned} \bullet 1 \cdot (1 \cdot 0) &= 1 \cdot 1 = 1 & \bullet 1 \cdot (1 \cdot 1) &= 1 \cdot 1 = 1 \\ (1 \cdot 1) \cdot 0 &= 1 \cdot 0 = 1 & (1 \cdot 1) \cdot 1 &= 1 \cdot 1 = 1 \end{aligned}$$

$$\begin{aligned} \bullet 1 \cdot (1 \cdot 2) &= 1 \cdot 2 = 2 & \bullet 1 \cdot (2 \cdot 0) &= 1 \cdot 2 = 2 \\ (1 \cdot 1) \cdot 2 &= 1 \cdot 2 = 2 & (1 \cdot 2) \cdot 0 &= 2 \cdot 0 = 2 \end{aligned}$$

$$\begin{aligned} \bullet 1 \cdot (2 \cdot 1) &= 1 \cdot 2 = 2 & \bullet 1 \cdot (2 \cdot 2) &= 1 \cdot 2 = 2 \\ (1 \cdot 2) \cdot 1 &= 2 \cdot 1 = 2 & (1 \cdot 2) \cdot 2 &= 2 \cdot 2 = 2 \end{aligned}$$

$$\begin{aligned} \bullet 2 \cdot (0 \cdot 0) &= 2 \cdot 0 = 2 & \bullet 2 \cdot (0 \cdot 1) &= 2 \cdot 1 = 2 \end{aligned}$$

$$\begin{aligned} \bullet 2 \cdot (0 \cdot 2) &= 2 \cdot 2 = 2 & \bullet 2 \cdot (1 \cdot 0) &= 2 \cdot 1 = 2 \\ (2 \cdot 0) \cdot 0 &= 2 \cdot 0 = 2 & (2 \cdot 0) \cdot 1 &= 2 \cdot 1 = 2 \end{aligned}$$

$$\begin{aligned} \bullet 2 \cdot (1 \cdot 0) &= 2 \cdot 1 = 2 & \bullet 2 \cdot (1 \cdot 1) &= 2 \cdot 0 = 2 \\ (2 \cdot 1) \cdot 0 &= 2 \cdot 0 = 2 & (2 \cdot 1) \cdot 1 &= 2 \cdot 1 = 2 \end{aligned}$$

$$\begin{aligned} \bullet 2 \cdot (1 \cdot 2) &= 2 \cdot 2 = 2 & \bullet 2 \cdot (2 \cdot 0) &= 2 \cdot 2 = 2 \\ (2 \cdot 1) \cdot 2 &= 2 \cdot 2 = 2 & (2 \cdot 2) \cdot 0 &= 2 \cdot 0 = 2 \end{aligned}$$

$$\begin{aligned} \bullet 2 \cdot (2 \cdot 1) &= 2 \cdot 2 = 2 & \bullet 2 \cdot (2 \cdot 2) &= 2 \cdot 2 = 2 \\ (2 \cdot 2) \cdot 1 &= 2 \cdot 1 = 2 & (2 \cdot 2) \cdot 2 &= 2 \cdot 2 = 2 \end{aligned}$$

$\therefore \cdot$ is an associative operator